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E-302 GENERATOR TRANSFORMERS

1. SCOPE OF WORKS

The work shall include the design, supply, manufacturing, factory testing, delivery to site, installation, commissioning and testing of generator transformers in accordance with Part 4, Subsection 1.2 Summary of Work.

The work shall also comply to general requirements for “Oil Immersed Power Transformers” detailed in the Technical Specification E-301.

2. CODES AND STANDARDS

The Contractor shall comply to the requirement of Part 4, Section 2, Subsection 2.5 “Codes and Standards” as well as specific codes and standards as mentioned in Technical Specification E-301.

3. OPERATING AND DESIGN CONDITIONS

The transformers shall be suitable for outdoor installation and capable to operate in power plant area with tropical sea coast.

The transformers shall be used in step-up services to connect steam turbine generator to the 500 kV substations. The circuit breakers (i.e. generator circuit breaker) shall be provided between each transformer and steam turbine generator.

The Bidder shall state any limitations with respect to the length of time that his transformers can operate satisfactorily at over voltage and/or over/under frequency conditions following such a trip-out.

The transformers shall be capable to operate in parallel operation with the other transformers.

4. DETAILED REQUIREMENTS

4.1 General

The transformers shall be three-phase, 50 Hz, step-up, two windings, outdoor, oil-

immersed, single tank, ONAN/ONAF type of cooling.

Surge arresters shall be furnished.

The 500 kV winding shall be graded per IEC Category 1.

Refer to the Technical Specification E-301 Subparagraph 3.4 for short circuit capability.

4.2 Transformer Ratings

	High Voltage Winding	Low Voltage Winding
- Nominal system voltage (line-to-line)..(kV)	500	Generator voltage
- Rated voltage transformer ratio	550 *)	Generator voltage
- Transformer connection.....(wye or delta)	Wye	Delta
- Rated winding voltage (line-to-neutral)(kV)	289	*)
- Insulation class	A	A
- Transformer rating full load at 65° C rise:		
a. Self cooling, ONAN.....(MVA)	*)	*)
b. Forced cooling, ONAF.....(MVA)	*)	*)
- Tap changing method : On-load tap changer		
a. Above rated voltage at full MVA capacity (number of taps and percent volts each tap)	8 x 1.25 %	None
b. Below rated voltage at full MVA capacity (Number of taps and percent volts each tap)	8 x 1.25 %	None
- System neutral (Isolated or type of grounding)	Solidly grounded	None

	High Voltage Winding	Low Voltage Winding	High Voltage Neutral
- System highest voltage(kV)	525	*)	*)
- Winding Basic Impulse Insulation Level (BIL)(kV _{peak})	1550	*)	*)

	High Voltage Winding	Low Voltage Winding	High Voltage Neutral
- Switching impulse withstand voltage(kV _{peak})	1175	--	--
- Power frequency withstand voltage(kV _{rms.})	620	*)	*)

NOTE: *)

- (1) Generator voltage shall be manufacturer's standard, see Technical Specification E-101 except otherwise specified.
- (2) Transformer ratio shall be verified by the Contractor during detailed design so that in all possible operating conditions of generator and transformer tap-positions the output voltage is between allowable 500 kV System voltage 475 kV - 525 kV for 500 kV Generator Transformer.
- (3) The MVA rating shall be determined by the Contractor and shall be equal to the highest possible continuous rating MVA of the Generator at the lowest possible rated continuous Generator Voltage.
- (4) The kV rating shall be equal to the highest possible voltage taking into account continuous and transient highest possible at Generator Voltage output.
- (5) All ratings calculation shall be submitted to the Owner for approval.
- (6) The Contractor shall be responsible of determining an impedance of transformers to meet the short circuit and voltage regulation requirement.

Others:

Angular displacement: The transformer shall have a standard phase relationship, vector group YNd1.

Impedance : The impedance at the rated connections shall have adequate value based on 100% rated MVA with tolerance IEC standard at 75°C.

Tap-changer :

A motor operated on-load tap changer shall be provided for operation when the transformer is fully loaded.

- (1) On-load tap changing equipment shall be suitable for supervisory control

and indication.

- (2) Remote at changing controls are required for mounting on the control desk in the Central Control Room.
- (3) Automatic voltage control equipment is required and shall be suitable for current compounding and provided with line drop compensation equipment.
- (4) Approximate size of tapping steps shall be 1.25 percent.
- (5) Excess of continuously applied voltage over tapping voltage for any tapping when transformer delivers its continuous rated power, shall be 5 %.
- (6) Tapping shall be in the high-voltage winding and shall be IEC category of voltage variation "CFVV".

4.3 Terminal Connections And Bushings

4.3.1 General

Transformer leads shall be brought out of the transformer case by means of bushings of the outdoor type. Bushings shall be capable of withstanding the 50 Hz applied voltage tests and lightning impulse tests specified.

4.3.2 Bushing ratings for 500 kV

	High Voltage Winding	Low Voltage Winding	High Voltage Neutral
- System highest voltage(kV)	525	Generator Voltage	--
- Basic impulse insulation level (BIL)(kV _{peak})	1,800	*)	150
- Creepage distance(mm)	16,275	*)	607
- Switching impulse withstand voltage(kV _{rms})	1,175	--	--
- Wet power frequency withstand voltage(kV _{rms})	680	*)	50

The color of the bushings shall be chocolate brown, wet process porcelain.

Bushings for the high voltage windings shall be suitable for connection to the 500 kV switchyard by means of overhead conductor. And also this bushings shall be suitable for connection to the 500 kV CGIB (Compressed Gas Insulated Bus) for steam turbine generator main circuit.

Bushing for the low voltage winding shall be top mounted and have a flanged enclosure suitable for connection to isolated phase bus duct. The low point of flanged enclosure shall be provided with a 15 mm diameter screened drain.

The high voltage neutral bushing shall be cover mounted.

The Contractor shall furnish and install a suitable copper bar between the high voltage neutral bushing on top of tank and ground pad on top of the tank.

4.4 Bushing Current Transformers

The Contractor shall design and provide current transformers for relaying and metering in accordance with the requirements of the Technical Specifications E-250, E-440, E-461 and on Bid drawing document.

The required current transformer burden ratings and other characteristics shall be determined by the Contractor and subject to Owner's approval.

4.5 Cooling System and Controls

The Contractor shall provide two (2) separate AC power supplies 400 Volt, three-phase, 50 Hz, 50 kA (short circuit) feed the cooling equipment, i.e. a normal and an alternative source. The DC supplies for the tripping circuits will be 220 volts DC nominal voltage, 20 kA (short circuits) with a voltage range of 188-242 volts.

An automatic transfer switch shall be provided to transfer the AC supply from normal to alternative source when the normal source is failed. The controller of cooling system shall be having the entire basic feature as described in Appendix A of Technical Specification E-301.

5. PERFORMANCE TESTING AND GUARANTEES

All control panels and control cabinets shall be completely assembled and tested in the factory. All tests shall be made in accordance with Technical Specification E-301 and the applicable IEC or ANSI/IEEE standards.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/control for the generator transformer shall be met the requirements of Part 4 Section 2 Subsection 2.7. The procedures as specified shall include, but not limited to, the following:

- (1) Design criteria and engineering.
- (2) Material control.
- (3) Factory and field inspections.
- (4) Control and function tests.
- (5) Wiring and insulation testing.
- (6) Insulation high voltage tests.
- (7) Protective relay and device settings and tests.
- (8) Welding.
- (9) Nondestructive testing.
- (10) Packaging and shipping.

The documentation as specified shall include, but not limited to, the following:

- (1) Inspection Reports.
- (2) Certified Test Reports.
- (3) Material Certifications.

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